

When Sitting Still Leads to a Standstill: The Role of Physical Activity and Sedentary Time in Coping with Academic Stress

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University students face considerable academic and psychosocial stressors, placing them at elevated risk for mental and physical health challenges compared to non-student peers (Sheldon et al., 2021). Physical activity (PA) has been shown to buffer stress and promote well-being in student populations, whereas sedentary time (ST) is linked to adverse outcomes such as anxiety and depressive symptoms (Wunsch et al., 2021). However, little is known about how PA and ST relate to broader patterns of academic behavior and experience which reflect students' personal experience with academic stress and the behavioral responses for coping with such stress. This study investigated the effect of PA and ST on transitions between academic behavior and experience patterns over time.

We used data from the Lübeck University Student Trial (LUST; Utesch et al., 2025), a longitudinal cohort survey study monitoring student health annually. The sample included 1,965 observations from $N = 785$ students ($M_{\text{age}} \pm \text{SD}$: 20.81 ± 3.33) enrolled between 2022 and 2025 who participated at least twice. Academic behavior and experience patterns were assessed using a student-adapted version of the Work-related Coping Behavior and Experience Patterns questionnaire (Schaarschmidt & Fischer, 1997), classifying respondents into health-promoting (G: Good health, S: Sparing investment) or health-endangering risk patterns (A: Ambitious, B: Burnout). PA and ST were measured using the short form of the International Physical Activity Questionnaire (Craig et al., 2003) and operationalized as metabolic equivalent of task minutes per week and as average daily sitting hours, respectively. Continuous-time multi-state Markov models were applied to estimate the effects of PA and ST on transition probabilities between patterns over time, modeling all patterns as transient states and allowing transitions between any pair of states.

Higher ST was associated with an increased hazard of transitioning from pattern G to pattern B within one year ($\text{HR} = 2.30$, 95% CI: 1.15 – 4.59). The probability of B being the subsequent pattern following G was 3.5% at low ST (-1 SD) and 16% at high ST (+1 SD). PA was not associated with transition hazards between patterns. The findings are preliminary and necessitate further validation due to a partially limited number of transition events.

The results suggest that reducing structurally mandated sitting may support more stable and health-promoting coping patterns in academic contexts. Universities could benefit from implementing active campus policies that integrate sedentism-reducing initiatives into campus life to strengthen students' coping capacity and prevent progression toward burnout-related coping.